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0. Executive Summary

This project develops and evaluates machine learning models to predict loan default risk using the Home Credit dataset, combining application data with multiple external behavioral and credit history sources. Extensive exploratory data analysis revealed strong class imbalance (~8% defaults), substantial missingness driven by structural absence of credit history, and predominantly weak linear correlations, highlighting the need for nonlinear models and carefully engineered features.

A comprehensive feature engineering pipeline was implemented across all datasets, including normalization through ratio-based financial features, behavioral aggregates from historical credit activity, anomaly handling, rare-category collapsing, and temporal filtering to emphasize recent behavior. All engineered features were merged into a single applicant-level dataset via left joins to preserve the original prediction population, resulting in a high-dimensional feature space of 386 features.

Three modeling approaches were evaluated: Logistic Regression as a linear baseline, and two gradient boosting models (XGBoost and LightGBM). Hyperparameter tuning was performed using Bayesian Optimization (Optuna), with computationally efficient strategies such as single-fold tuning and explicit regularization. Feature selection was applied for tree-based models by removing the bottom 30% of least-informative features, reducing the final feature set to 272 columns, and improving stability and efficiency.

Results show that both XGBoost and LightGBM substantially outperform the linear baseline, achieving validation AUCs around 0.79 and PR-AUCs around 0.28, while maintaining high recall for the minority (default) class, which is an intentional and appropriate trade-off for credit risk applications. Feature importance analyses consistently highlight external credit scores, repayment behavior, utilization ratios, and delinquency signals as the strongest predictors, aligning with earlier correlation findings and domain expectations.

While the models achieve competitive performance, they do not surpass top competition results, largely due to limited experimentation time, constrained hyperparameter search, and the absence of advanced ensembling or deep temporal modeling. Nonetheless, the project demonstrates a robust, principled, and end-to-end credit risk modeling framework that effectively integrates heterogeneous data sources and balances predictive performance with interpretability and computational feasibility.

1. Application Data (Train/Test)

The Application dataset contains the core information available at the time a loan application is submitted. Each row corresponds to a single applicant and includes demographic characteristics, income and employment details, requested loan terms, and basic housing and contact information. This dataset also contains the target variable, indicating whether the applicant later defaulted on the loan, making it the primary source for modeling credit risk in this project.

1.1. EDA Highlights

The Application dataset contains approximately 300,000 loan applications with a little over 120 features per applicant. A notable characteristic of this dataset is the presence of missing values: around 60 columns contain missing data, with some features missing values for up to 70% of applicants. Importantly, this missingness is not uniformly distributed and often reflects real-world conditions rather than data quality issues.

1.1.1. Categorical Variables

CODE_GENDER

The dataset contains a higher proportion of female applicants than male applicants. Approximately 65% of applications are from women, while 35% are from men. This imbalance reflects the applicant pool rather than model bias, but should be monitored during modeling and evaluation.

ORGANIZATION_TYPE

The variable contains 58 categories, indicating high cardinality. However, its distribution is relatively spread out, with most categories representing small but comparable proportions of applicants and no single category dominating. As a result, collapsing categories based on frequency would likely lead to a loss of meaningful employer-related information. The variable was therefore retained in its original form and handled carefully during encoding.

NAME_CONTRACT_TYPE

Most applications correspond to “Cash loans”, which account for roughly 90% of all applications, while “Revolving loans” make up the remaining share. This strong dominance suggests that contract type alone may have limited standalone predictive power but may still interact with other financial features.

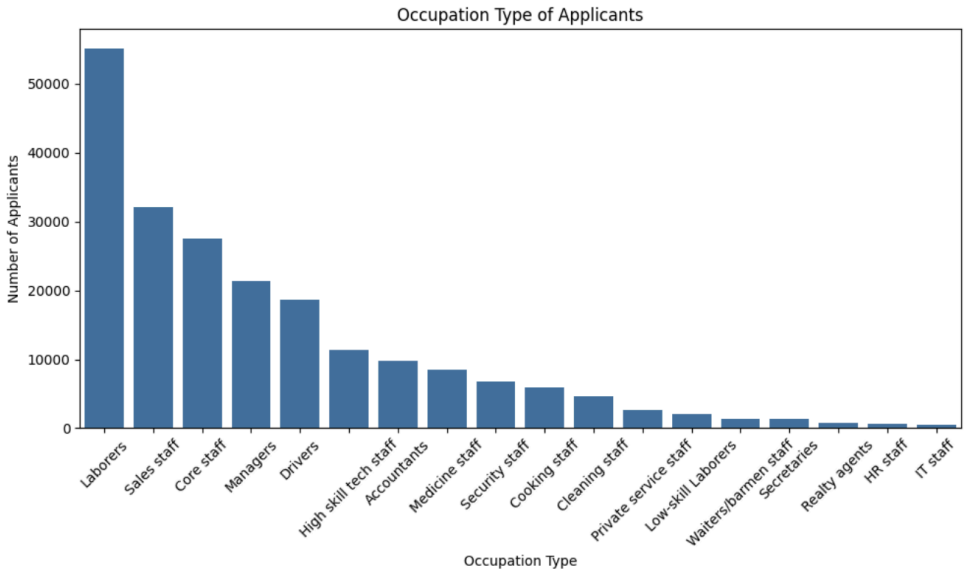
NAME_INCOME_TYPE

NAME_INCOME_TYPE	proportion
Working	0.516320
Commercial associate	0.232892
Pensioner	0.180033
State servant	0.070576
Unemployed	0.000072
Student	0.000059
Businessman	0.000033
Maternity leave	0.000016

This variable includes a wide range of income categories with highly uneven representation. While categories such as “Working” and “Commercial Associate” dominate, several categories appear with very small proportions, including “Businessman”, “Student”, “Unemployed”, “Maternity Leave”, and “Pensioner”. The presence of many low-frequency categories suggests that category consolidation may be necessary to reduce noise and improve model stability.

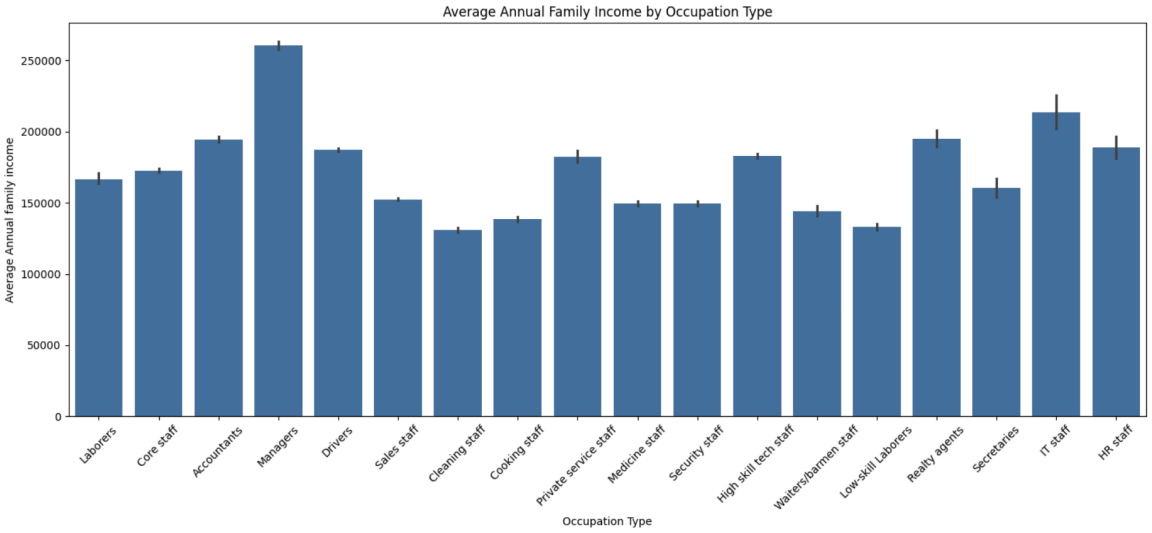
OCCUPATION_TYPE

Occupation type contains many categories and a high proportion of missing values. The distribution across occupations is uneven, and missingness itself may reflect informal employment or incomplete reporting. Due to this complexity, this variable requires both careful visualization and thoughtful encoding.



OCCUPATION_TYPE + AMT_INCOME_TOTAL

Average income varies substantially across occupation types, with higher-skilled roles such as “Managers”, “IT staff”, “Accountants”, and “HR staff” earning significantly more than low-skill labor, cleaning, cooking, and service roles. This clear separation confirms that occupation type reflects meaningful differences in earning capacity, though variability within groups suggests it should be interpreted in combination with other financial features.

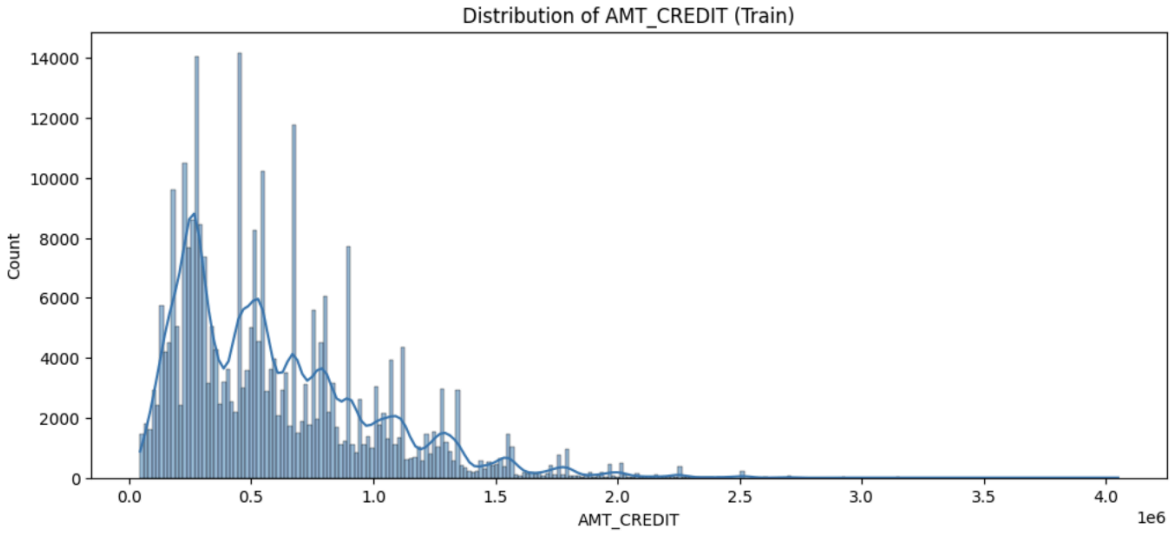


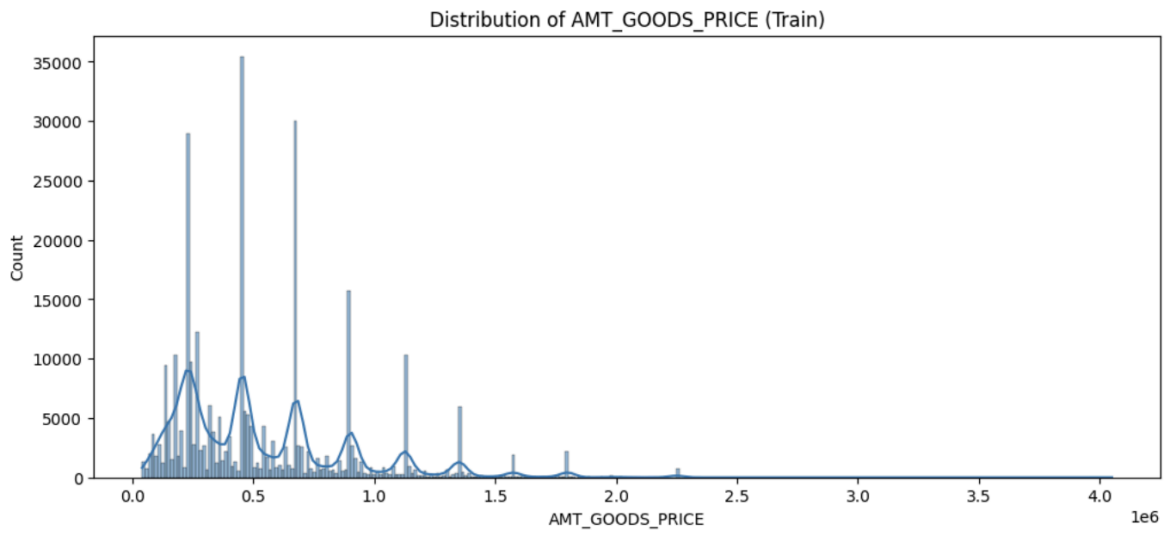
CNT_CHILDREN

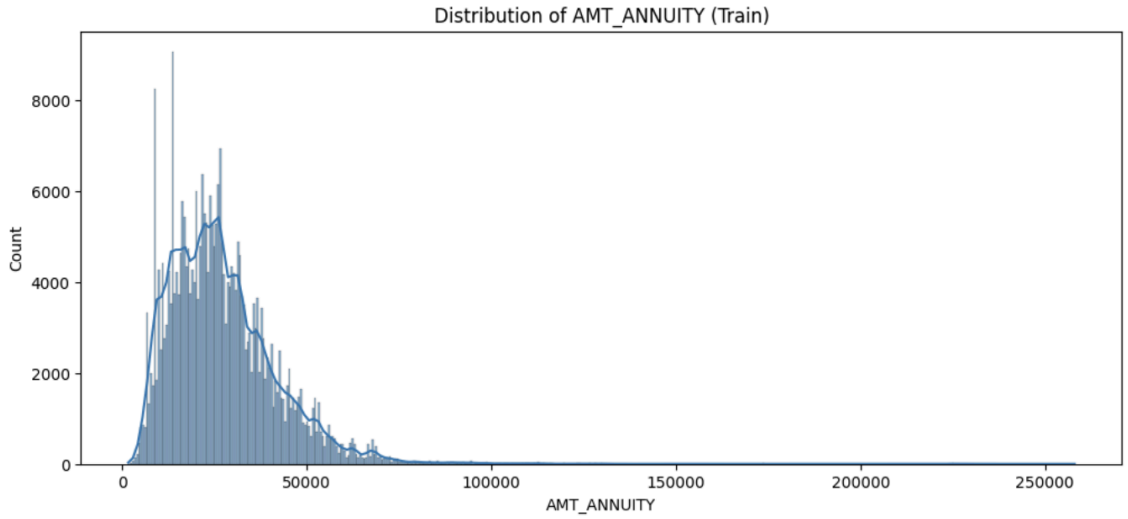
The majority of applicants report zero to two children, while a small number report six or more, with values reaching up to 19 children. These extreme cases are atypical and may represent outliers or special household structures, warranting careful treatment to avoid disproportionate influence on the model

1.1.2. Numerical Variables

Many numerical variables in the Application dataset exhibit strong right skew, particularly financial features such as income, credit amount, and annuity. This indicates substantial heterogeneity among applicants, with a small number of individuals having disproportionately large values. As a result, raw monetary amounts alone are not always directly comparable across applicants and require careful treatment in subsequent modeling steps.







Several time-related variables, including age and employment duration, are recorded as negative values because they are measured as the number of days relative to the application date. While unintuitive at first glance, this representation is consistent across the dataset and does not pose a problem as long as the relative ordering and scale are preserved.

DAYS_EMPLOYED

The variable exhibits notable anomalies, with a subset of applicants having extreme positive values (365243 days) that do not correspond to realistic employment durations. These values likely represent placeholder codes rather than true employment histories and must be handled explicitly to prevent distortion of downstream features.

OWN_CAR_AGE

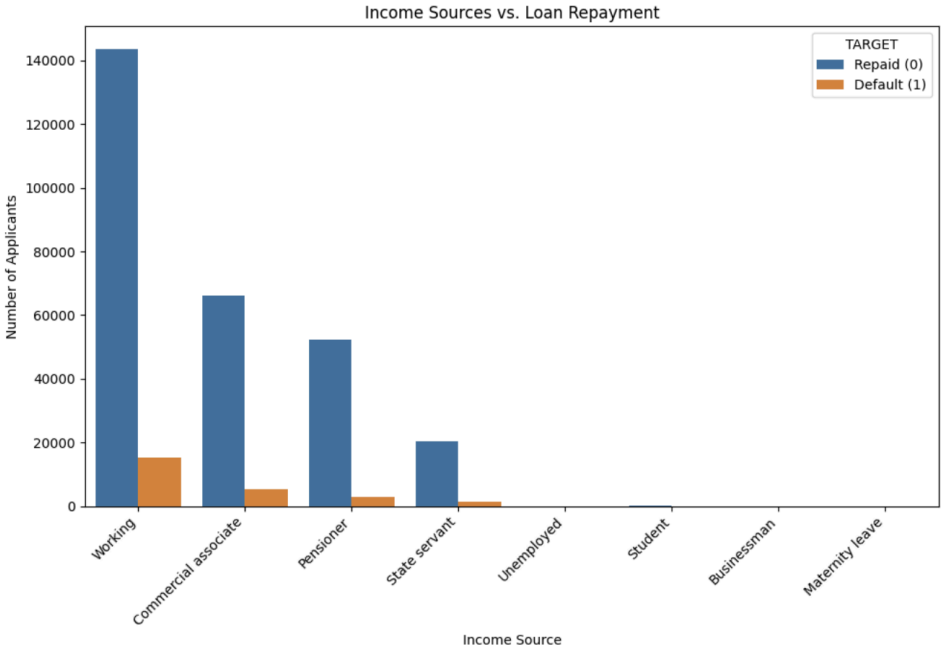
The variable shows the presence of outliers, with unusually large values relative to typical car ownership durations (91 years). These extreme cases are rare but can disproportionately influence model behavior if left untreated, motivating additional inspection and potential capping or transformation.

1.1.3. Target Variable

The target variable, indicating whether an applicant defaulted, is highly imbalanced. Approximately 8% of applications result in default, while 92% do not, reflecting the real-world nature of consumer lending where defaults are relatively rare. This imbalance has important implications for model evaluation, as metrics such as accuracy alone would be misleading.

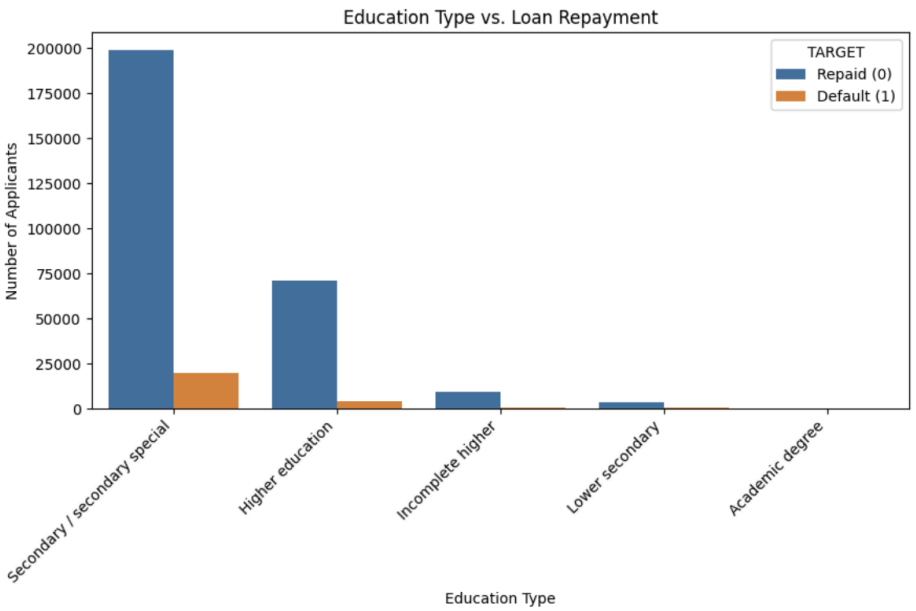
TARGET + NAME_INCOME_TYPE

The plot shows that most applicants come from stable income sources such as Working, Commercial Associate, and Pensioner, which together account for the majority of both repayments and defaults. While these groups also contribute the largest number of defaults in absolute terms, this primarily reflects their higher representation in the applicant pool rather than elevated individual risk. In contrast, income sources with very small sample sizes (such as Student, Businessman, Maternity Leave) contribute minimally to overall outcomes, making their default behavior less reliable to interpret and highlighting the need for cautious treatment of low-frequency categories.



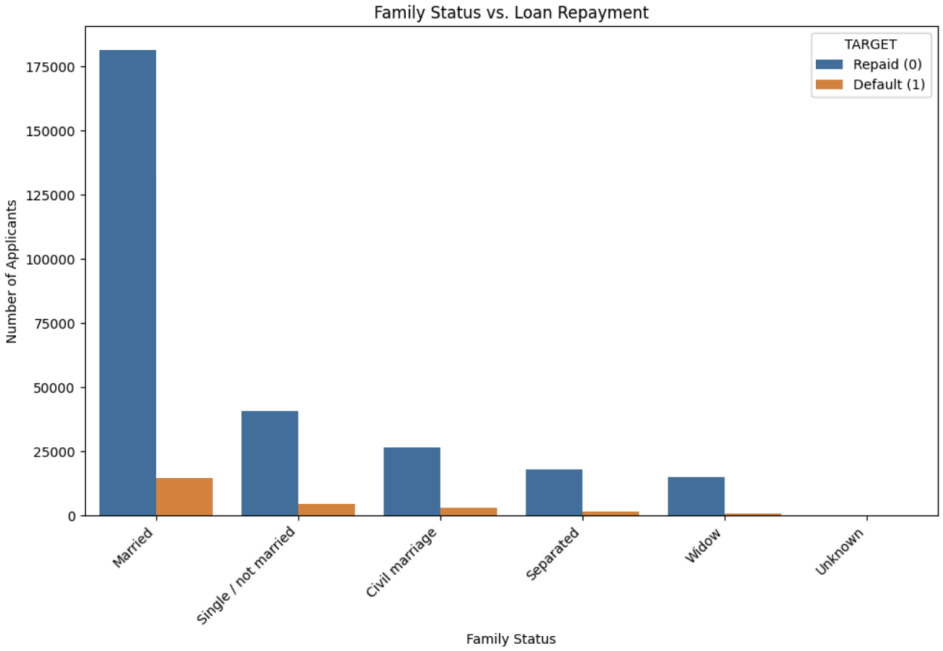
TARGET + NAME_EDUCATION_TYPE

Most applicants have secondary or secondary special education, followed by those with higher education, making these groups the primary contributors to both repayments and defaults in absolute terms. Applicants with higher levels of education generally show fewer defaults relative to their population size, while categories with very small representation (such as academic degree or lower secondary) contribute little reliable signal due to limited sample size.



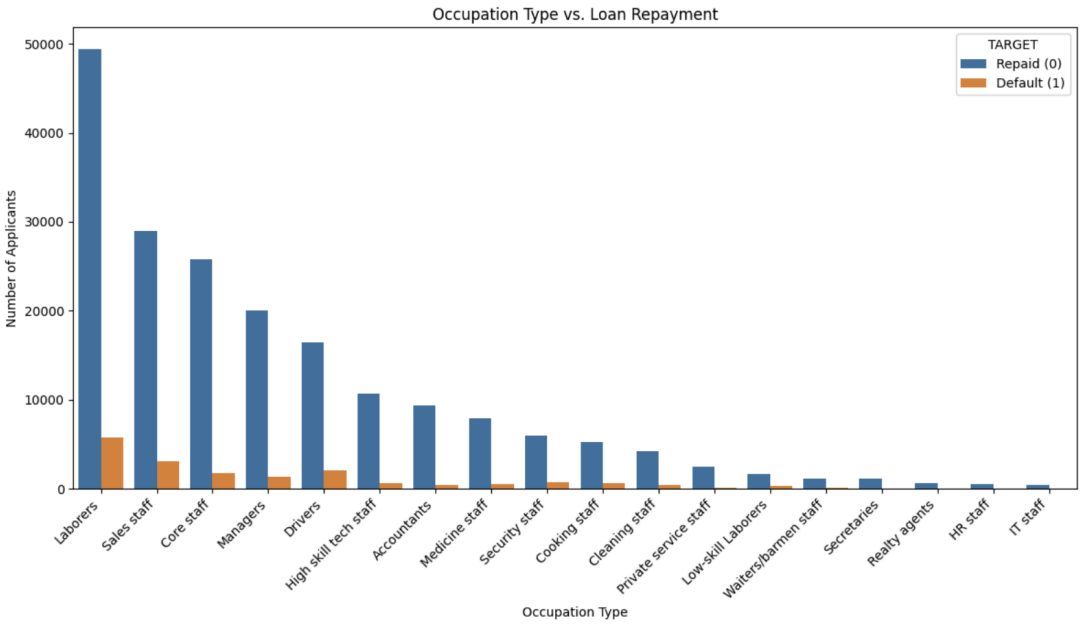
TARGET + NAME_FAMILY_STATUS

Most applicants are married, followed by single/not married, making these groups the largest contributors to both repayments and defaults in absolute terms. While differences in default counts largely reflect group size, smaller categories such as separated and widow show fewer observations, limiting the reliability of standalone conclusions.



TARGET + OCCUPATION_TYPE

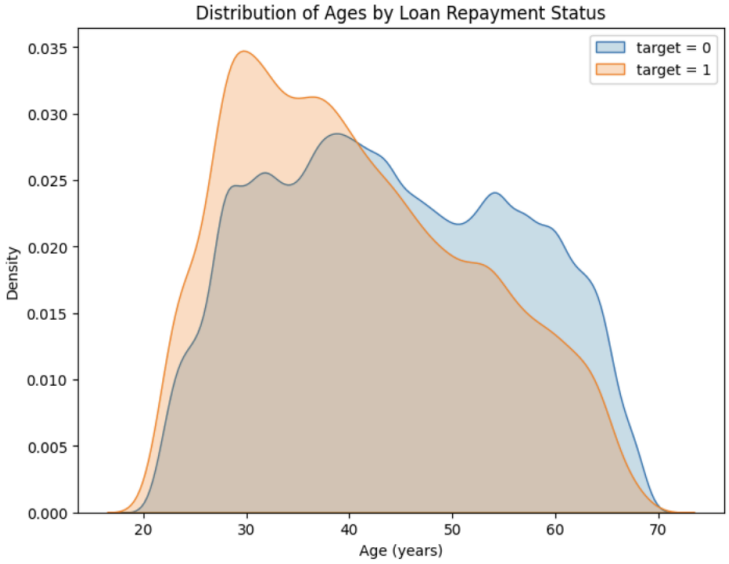
Applicants are concentrated in a small number of occupation groups, particularly laborers, sales staff, core staff, and managers, which account for the majority of both repayments and defaults. Occupations with larger applicant counts naturally show higher absolute default numbers, while many other occupations appear infrequently, limiting the reliability of default comparisons for those groups.

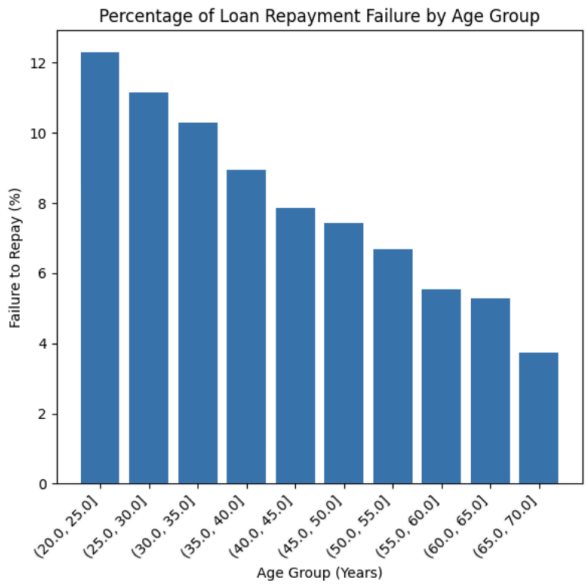


TARGET + DAYS_BIRTH

The age distribution shows a clear shift between applicants who repay and those who default. Younger applicants are overrepresented among defaulters, while older applicants are more prevalent among successful repayments. This pattern becomes more explicit when examining default rates by age group:

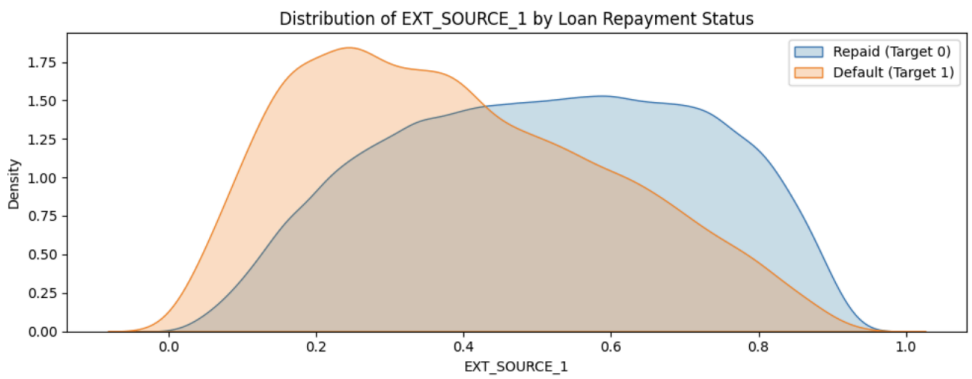
the likelihood of default steadily decreases with age, dropping from above 12% in the youngest group (20–25) to below 4% for applicants aged 65–70.

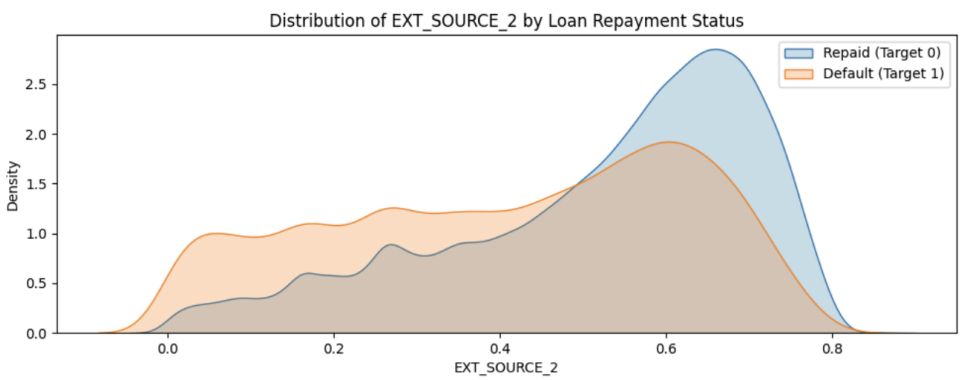


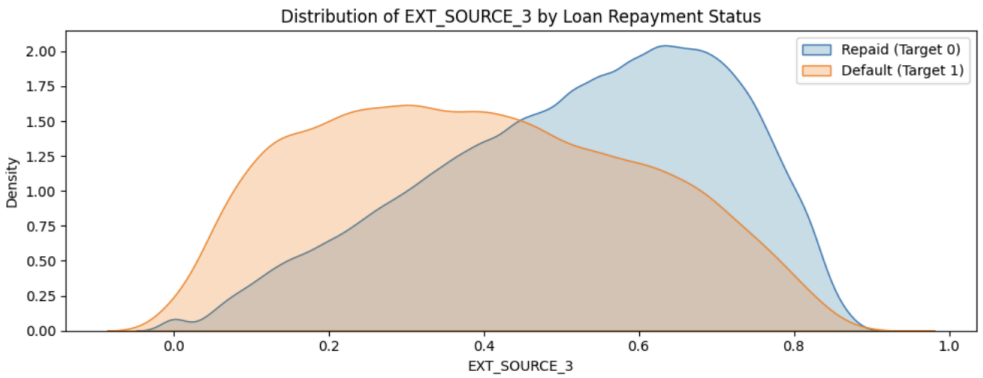


TARGET + EXT_SOURCE_1/2/3

The distributions of EXT_SOURCE_1, EXT_SOURCE_2, and EXT_SOURCE_3 show a clear and consistent separation between applicants who repay and those who default. Defaulters are heavily concentrated at lower score values, while successful repayments are shifted toward higher scores across all three sources. This pattern is monotonic and consistent, indicating that higher external scores are strongly associated with lower default risk.







1.1.4. Correlation with Target Variable

Most Positive Correlations:

	TARGET
FLAG_WORK_PHONE	0.028524
DEF_60_CNT_SOCIAL_CIRCLE	0.031276
DEF_30_CNT_SOCIAL_CIRCLE	0.032248
LIVE_CITY_NOT_WORK_CITY	0.032518
OWN_CAR_AGE	0.037612
DAYS_REGISTRATION	0.041975
FLAG_DOCUMENT_3	0.044346
REG_CITY_NOT_LIVE_CITY	0.044395
FLAG_EMP_PHONE	0.045982
REG_CITY_NOT_WORK_CITY	0.050994
DAYS_ID_PUBLISH	0.051457
DAYS_LAST_PHONE_CHANGE	0.055218
REGION_RATING_CLIENT	0.058899
REGION_RATING_CLIENT_W_CITY	0.060893
DAYS_BIRTH	0.078239

Most Negative Correlations:

	TARGET
EXT_SOURCE_3	-0.178919
EXT_SOURCE_2	-0.160472
EXT_SOURCE_1	-0.155317
DAYS_EMPLOYED	-0.044932
FLOORSMAX_AVG	-0.044003
FLOORSMAX_MEDI	-0.043768
FLOORSMAX_MODE	-0.043226
AMT_GOODS_PRICE	-0.039645
REGION_POPULATION_RELATIVE	-0.037227
ELEVATORS_AVG	-0.034199
ELEVATORS_MEDI	-0.033863
FLOORSMIN_AVG	-0.033614
FLOORSMIN_MEDI	-0.033394
LIVINGAREA_AVG	-0.032997
LIVINGAREA_MEDI	-0.032739

Simple correlation analysis shows that most individual Application features have weak linear correlation with the target, which is expected in credit risk problems driven by nonlinear interactions. The strongest negative correlations are observed for the external source scores (EXT_SOURCE_1/2/3), confirming their strong inverse relationship with default risk. Age (DAYS_BIRTH) and employment-related variables also show modest correlation, with older and more stably employed applicants tending to default less.

1.2. Feature Engineering

Feature engineering for the Application dataset was designed to improve signal quality, reduce noise and sparsity, and translate raw applicant information into features that better reflect credit risk. To ensure consistency between training and test data, the two datasets were first combined, and all transformations were applied jointly. This avoids discrepancies in encoding, scaling, and feature availability across splits.

1.2.1. Data Cleaning and Anomaly Handling

Initial cleaning focused on removing clearly invalid or inconsistent records and addressing known data anomalies. Applications with undefined gender values (CODE_GENDER) or unknown family status (NAME_FAMILY_STATUS) were excluded, as these categories do not carry meaningful information and appear infrequently. A well-known anomaly in the dataset, where DAYS_EMPLOYED takes an unrealistically large placeholder value, was replaced with missing values to prevent distortion of employment-related features.

Extreme values in select variables were capped to reduce the influence of outliers while preserving relative ordering. For example, car ownership age (OWN_CAR_AGE) was capped at a realistic upper bound, and the number of reported children (CNT_CHILDREN) was limited to a small maximum, as very large values are rare and potentially unreliable.

Highly skewed monetary variables such as income (AMT_INCOME_TOTAL), credit amount (AMT_CREDIT), annuity (AMT_ANNUITY), and goods price (AMT_GOODS_PRICE) were log-transformed. This reduces the impact of extreme values, improves numerical stability, and allows models to focus on relative differences rather than absolute scale.

A predefined list of columns was dropped to reduce noise and redundancy. These primarily include:

- Highly missing building and housing attributes.
- Sparse categorical housing descriptors.
- Redundant document and contact flags.
- Strongly correlated variants of the same underlying measurement.

1.2.2. New Features

A. Ratio-Based Financial Features – several ratio features were created to normalize raw monetary values and make them comparable across applicants with different scales. These ratios are more informative than absolute values alone because they directly represent affordability and financial pressure:

R_INCOME_TO_CREDIT = AMT_INCOME_TOTAL / AMT_CREDIT: Measures how large the requested loan is relative to total income, serving as a direct proxy for affordability.
R_ANNUITY_TO_CREDIT = AMT_ANNUITY / AMT_CREDIT: Captures repayment burden by relating periodic payments to total loan size.
R_CREDIT_TO_GOODS = AMT_CREDIT / AMT_GOODS_PRICE: Highlights cases where the loan amount exceeds the declared price of goods, which may indicate higher risk or additional borrowing needs.
INCOME_PER_PERSON = AMT_INCOME_TOTAL / CNT_FAM_MEMBERS: Adjusts household income by the number of family members, providing a more realistic measure of disposable income.
FLAG_CREDIT_GT_GOODS: Binary indicator flagging whether the requested credit exceeds the goods price.

B. Employment and Stability Indicators

R_EMPLOYED_TO_AGE: Normalizes employment duration by applicant age, capturing employment stability relative to life stage rather than absolute tenure. This feature helps distinguish between naturally short employment histories for younger applicants and potential instability for older applicants.

C. Aggregated Behavioral and Coverage Signals – to reduce sparsity and summarize related indicators, several aggregate features were introduced:

CONTACT_SCORE: Aggregates multiple contact-related flags (FLAG_MOBIL, FLAG_EMP_PHONE, FLAG_WORK_PHONE, FLAG_CONT_MOBILE, FLAG_EMAIL) into a single measure of contactability and data completeness.
DOC_MISS_COUNT: Counts the number of missing document flags across all required documents.
FLAG_DOC_MISSING: Binary indicator capturing whether any required documentation is missing.
BUREAU_REQ_TOTAL: Aggregates recent credit bureau request counts into a single measure of recent credit-seeking activity.

1.2.3. Categorical Encoding

Categorical features were handled with care to balance information preservation and sparsity reduction. Rare categories in selected variables (NAME_INCOME_TYPE) were grouped to reduce noise, while other high-cardinality variables were left intact where distributions were sufficiently spread out.

Binary categorical variables without missing values were label-encoded, while remaining categorical variables were one-hot encoded. Missing values were explicitly treated as a separate category where appropriate, allowing the model to learn from missingness itself when it carries information.

2. Bureau Data

The Bureau dataset contains historical credit information for applicants obtained from external credit bureaus. Each row corresponds to a single prior or active credit account linked to an applicant and includes details such as credit type, loan amount, outstanding debt, repayment status, and key dates related to the credit lifecycle.

2.1. EDA Highlights

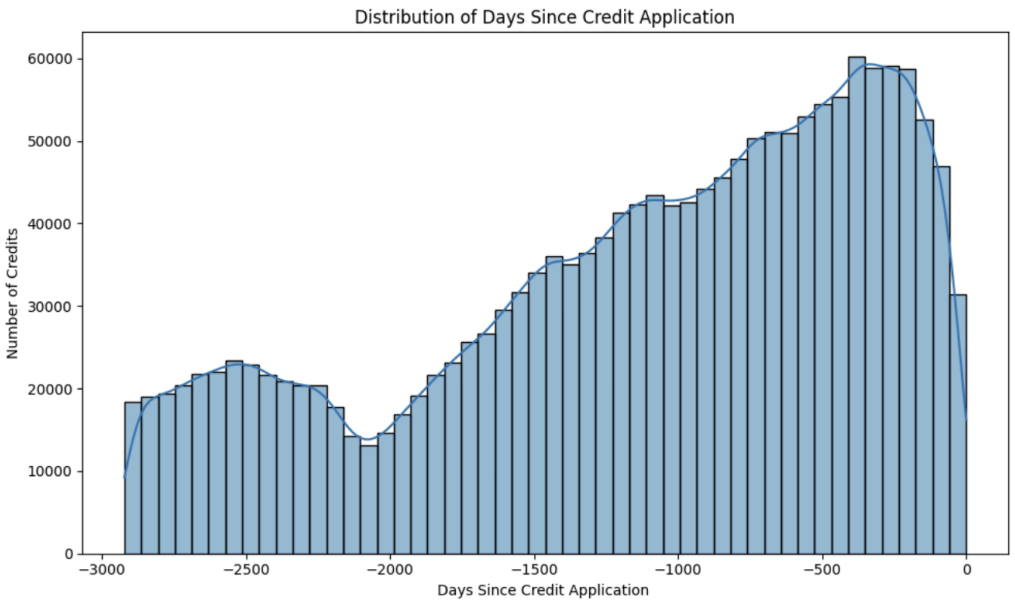
The Bureau dataset contains approximately 1.7 million historical credit records, corresponding to about 305,000 unique applicants, with each applicant potentially associated with multiple bureau loans. The dataset includes numerical variables describing credit amounts, outstanding debt, and loan timelines, as well as categorical variables indicating credit type, credit currency, and credit status. Several financial and date-related features contain missing values, with missingness exceeding 60–70% for some columns.

2.1.1. Categorical Variables

The categorical variables in the Bureau dataset are strongly imbalanced. CREDIT_CURRENCY is almost entirely dominated by a single currency (over 99% of records), with all other currencies appearing only sporadically. CREDIT_ACTIVE is primarily split between Active and Closed credits (together accounting for around 95% of records), while other statuses are rare. Similarly, CREDIT_TYPE is dominated by a small number of loan types, with the top categories covering the majority of records and a long tail of infrequent types. Due to this imbalance and sparsity, rare categories should be collapsed to reduce noise and prevent unnecessary feature expansion during encoding.

2.1.2. Numerical Variables

The distribution of days since credit application (DAYS_CREDIT) shows that most bureau records correspond to recent credit activity, with a strong concentration closer to the application date and a long tail representing older credit accounts. This pattern indicates that applicants often have a mix of recent and historical credit records, allowing the dataset to capture both short-term borrowing behavior and longer credit history. The smooth, continuous shape of the distribution suggests no obvious data anomalies, and the wide range of values highlights the importance of recency when summarizing bureau information for modeling.



2.2. Feature Engineering

The Bureau dataset contains multiple credit records per applicant, so feature engineering focuses on summarizing each applicant's overall credit portfolio and attaching these summaries to every bureau loan record. Applicant-level statistics are computed by grouping records by SK_ID_CURR and then broadcast back to the loan level, ensuring that each credit record is enriched with context about the applicant's broader credit history.

2.2.1. New Features

A. Credit History Size and Diversity – summarize the scale and variety of an applicant's credit history.

NEW_LOAN_COUNT: Captures the total number of bureau loans associated with the applicant.
NEW_CREDIT_TYPE_NUNIQUE: Measures the number of distinct credit types held.

B. Credit Activity and Recency – describes how much of an applicant's credit portfolio is currently active.

NEW_ACTIVE_SHARE: Represents the share of loans that are active.
NEW_FUTURE_ENDDATE_SHARE: Captures the proportion of loans with end dates in the future.

C. Debt Burden and Leverage – provides a normalized view of an applicant's remaining repayment burden.

NEW_DEBT_TO_CREDIT: Measures the ratio of total outstanding debt to total credited amount across all bureau loans.

3. Bureau Balance Data

The Bureau Balance dataset provides monthly snapshots of the repayment status for each bureau-recorded credit, capturing how loan balances evolve over time. Each record corresponds to a specific loan (SK_ID_BUREAU) at a given month, with the dataset covering approximately 817,000 unique bureau loans. The dataset includes information on payment status, indicating whether installments were paid on time, overdue, or in default, and adds a temporal perspective to applicants' credit behavior.

3.1. EDA Highlights

STATUS	count
C	13646993
0	7499507
X	5810482
1	242347
5	62406
2	23419
3	8924
4	5847

The primary variable of interest is STATUS, which captures repayment behavior ranging from on-time payments to severe delinquency. Most records correspond to non-delinquent states (C, 0, X), while delinquent states are much less frequent. Mild delinquency (1–2) occurs relatively rarely, and severe delinquency (3–5) is observed in only a small fraction of records.

3.2. Feature Engineering

Feature engineering for the Bureau Balance dataset focuses on summarizing monthly repayment behavior at the loan level. Since each bureau loan (SK_ID_BUREAU) can have multiple monthly status records, features are constructed by aggregating these monthly observations into compact indicators that describe repayment history, coverage, and delinquency patterns for each loan.

3.2.1. New Features

A. Repayment History Length and Coverage – to capture how much historical information is available for each loan, several time-based features are constructed from MONTHS_BALANCE:

BB_MONTHS_MIN: earliest month recorded for the loan.
BB_MONTHS_MAX: most recent month recorded.
BB_MONTHS_COUNT: total number of monthly records available.

B. Delinquency and Status Proportions – the categorical repayment status variable (STATUS) is first one-hot encoded. Then, for each repayment status, a proportion-based feature is created:

BB_STATUS_*_MEAN (ex: BB_STATUS_1_MEAN, BB_STATUS_5_MEAN): Each feature represents the fraction of months a loan spent in a specific repayment state.

3.3. Bureau + Bureau Balance

3.3.1. Loan-level Enrichment (SK_ID_BUREAU)

After feature engineering the Bureau and Bureau Balance datasets separately, the two sources are then merged at the loan level using SK_ID_BUREAU. At this stage, categorical variables are processed by grouping rare categories and applying one-hot encoding, ensuring a compact and stable numerical representation.

3.3.2. Aggregation to Applicant-level (SK_ID_CURR)

The merged loan-level dataset is then aggregated to the applicant level (SK_ID_CURR) to construct the final Bureau feature table. Different aggregation strategies are applied depending on feature type:

- *Numerical aggregations:* Minimum and maximum values are used for time-related variables to capture the span of credit history, while mean, sum, or maximum statistics are applied to credit amounts, debt, and count-based variables to summarize overall exposure and risk.
- *Categorical aggregations:* Categorical variables are aggregated using means.

4. Previous Applications Data

The Previous Applications dataset contains information on loan applications previously submitted by applicants to Home Credit. Each row corresponds to a single past application and includes details such as the application decision, requested and approved loan amounts, contract type, and key dates related to the application process.

4.1. EDA Highlights

The Previous Applications dataset contains approximately 1.67 million historical application records across 37 features, linked to about 339,000 unique applicants, with each applicant potentially associated with multiple prior applications. The dataset exhibits substantial missingness across several variables, particularly interest rate–related and payment timing features, where missing values exceed 50% and reach nearly 100% for some columns.

4.1.1. Categorical Variables

The Previous Applications dataset contains categorical variables with varying cardinality. A small number of features exhibit moderate to high cardinality, with the largest having around 25–30 unique categories (NAME_GOODS_CATEGORY, NAME_CASH_LOAN_PURPOSE), which increases the risk of sparse encodings.
Separately, several categorical variables display highly skewed distributions (NAME_GOODS_CATEGORY, NAME_SELLER_INDUSTRY, NAME_CASH_LOAN_PURPOSE, CHANNEL_TYPE), where a few dominant categories account for most observations, and many others appear infrequently. These variables require category collapsing to reduce sparsity and improve robustness before encoding.

4.1.2. Numerical Variables

Some time-related numerical features contain anomalous placeholder values. In particular, DAYS_FIRST_DRAWING equals 365243 for the majority of observations (around 934k rows), which does not represent a real date but instead indicates that the event did not occur. Valid entries are negative and measured relative to the application date. This value should be treated as a placeholder rather than a true numerical value during preprocessing.

4.2. Feature Engineering

Feature engineering is first performed at the previous-application (loan) level, where each row corresponds to a single historical loan application linked to an applicant, and the resulting features are subsequently aggregated to the applicant level to summarize overall prior borrowing behavior.

4.2.1. Data Cleaning and Anomaly Handling

- Columns with near-complete missingness (RATE_INTEREST_PRIMARY, RATE_INTEREST_PRIVILEGED) are dropped.

- Known placeholder values (365243) in time-related columns (DAYS_FIRST_DRAWING, DAYS_TERMINATION, DAYS_LAST_DUE_1ST_VERSION, DAYS_FIRST_DUE, DAYS_LAST_DUE) are replaced with NaN to avoid introducing artificial signals.
- Weekday information is simplified by collapsing Monday–Friday to WEEKDAY and Saturday–Sunday to WEEKEND, reducing categorical sparsity while preserving temporal meaning.

4.2.2. New Features

A. Ratio-Based Financial Features – to better capture loan structure and affordability, several scale-invariant ratio features are created:

NEW_APP_CREDIT_RATE = AMT_APPLICATION / AMT_CREDIT: Measures how much of the approved credit was actually requested.
NEW_APP_PAYMENT_RATE = AMT_CREDIT / AMT_ANNUITY: Proxy for repayment burden per installment.
NEW_APP_GOODS_RATE = AMT_APPLICATION / AMT_GOODS_PRICE: Indicates alignment between the requested amount and the goods value.
NEW_CREDIT_GOODS_RATE = AMT_CREDIT / AMT_GOODS_PRICE: Captures over- or under-financing of the purchase.

B. Interest Proxies – since explicit interest rates are largely unavailable, interest is approximated using the payment structure:

NEW_INTEREST = CNT_PAYMENT × AMT_ANNUITY − AMT_CREDIT**:** Approximate total interest paid over the loan.
NEW_INTEREST_RATE: Normalized interest proxy accounting for loan size and duration.
NEW_INTEREST_SHARE = NEW_INTEREST / AMT_CREDIT: Relative interest burden.

C. Time-Delta Features – captures deviations between expected and actual loan termination, which may reflect early repayment or irregular closure.
NEW_END_DIFF = DAYS_TERMINATION − DAYS_LAST_DUE

4.2.3. Categorical Encoding

Before aggregation, categorical variables are prepared to ensure stable downstream statistics:

- **Rare category collapsing** is applied to skewed, high-cardinality columns (NAME_TYPE_SUITE, NAME_GOODS_CATEGORY, CHANNEL_TYPE, NAME_SELLER_INDUSTRY, NAME_CASH_LOAN_PURPOSE) to reduce one-hot sparsity.
- Binary categorical variables (two levels, no missing values) are **label-encoded**.
- Remaining categorical variables are **one-hot encoded**, producing interpretable dummy features suitable for aggregation.

4.2.4. Aggregation to Applicant Level (SK_ID_CURR)

After loan-level feature construction and encoding, all previous applications are aggregated to the applicant level.

- *Numerical Aggregations:* Minimum, maximum, and mean statistics are applied to monetary variables (credit amount, annuity, goods price) to capture typical and extreme borrowing behavior across past applications. Count-based features summarize application frequency, while engineered ratios, interest proxies, and time-delta features are aggregated using mean and max values to reflect both average conditions and worst-case loan terms at the applicant level.
- *Categorical Aggregations:* One-hot encoded categorical features are aggregated using the mean.

5. POS CASH Balance Data

The POS CASH Balance dataset captures monthly payment behavior for point-of-sale and cash loans previously issued to applicants. Each record corresponds to a specific loan (SK_ID_PREV) at a given month and includes information on installment status, remaining payments, and delinquency indicators.

5.1. EDA Highlights

The POS CASH Balance dataset contains approximately 10 million monthly records across 8 features, covering repayment histories for approximately 337,000 applicants. Contract status (NAME_CONTRACT_STATUS) is strongly imbalanced, with the vast majority of records marked as Active (around 91.5%), followed by Completed (around 7.4%), while all other statuses each account for less than 1% of observations.

5.2. Feature Engineering

5.2.1. Temporal Filtering

POS CASH records first are restricted to a fixed recent window by retaining only observations from the last 12 months (MONTHS_BALANCE ≥ −12). This focuses the features on recent repayment behavior, which is more informative for credit risk modeling, while also reducing noise from older history.

5.2.2. Categorical Encoding

The categorical variable NAME_CONTRACT_STATUS is highly imbalanced, with most records labeled as Active. To reduce sparsity, **rare status categories are collapsed** before encoding. All categorical variables are then **one-hot encoded**.

5.2.3. Aggregation to Applicant Level (SK_ID_CURR)

After monthly POS records are processed and encoded, all observations are aggregated to the applicant level.

- *Numerical Aggregations:* Timeline features (MONTHS_BALANCE) are summarized using maximum, mean, and count statistics to capture recency, average timing, and the length of observed POS activity. Delinquency indicators (SK_DPD, SK_DPD_DEF) are aggregated using maximum and mean values to reflect both worst-case and typical repayment behavior over the recent history window.
- *Categorical Aggregations:* One-hot encoded contract status features are aggregated using the mean.

6. Installment Payments Data

The Installments Payments dataset records detailed repayment information for individual loan installments associated with applicants. Each row corresponds to a single installment payment and

includes the scheduled payment amount and date, the actual payment made, and timing differences between expected and actual payments.

6.1. EDA Highlights

The Installments Payments dataset contains approximately 13.6 million installment records across 8 variables, covering repayment activity for approximately 339,600 applicants. The installment version indicator (NUM_INSTALMENT_VERSION) is highly skewed, with the majority of records concentrated in a small number of versions (most commonly versions 0 and 1), while a long tail of rare versions appears only a handful of times.

6.2. Feature Engineering

6.2.1. Temporal Filtering

Installment records are first restricted to recent payment activity by keeping only installments scheduled within the last 365 days prior to application (DAYS_INSTALMENT ≥ −365). This focuses the feature set on the most relevant repayment behavior while reducing noise from older installment history.

6.2.2. New Features

Several features are constructed at the installment level to capture payment adequacy and timing:

PAYMENT_PERC = AMT_PAYMENT / AMT_INSTALMENT: Ratio of the actual payment to the scheduled installment amount, indicating underpayment or overpayment.
PAYMENT_DIFF = AMT_INSTALMENT – AMT_PAYMENT: Difference between the scheduled installment and the actual payment, capturing payment shortfalls or excess payments.
Late payment days (DPD) = DAYS_ENTRY_PAYMENT – DAYS_INSTALMENT: Measures payment delays.
Early payment days (DBD) = DAYS_INSTALMENT – DAYS_ENTRY_PAYMENT: Captures early repayments.

6.2.3. Aggregation to Applicant Level (SK_ID_CURR)

After row-level feature construction, all installment records are aggregated to the applicant level. Numerical features are summarized using maximum, mean, and sum statistics to capture worst-case, typical, and cumulative repayment behavior, while count-based features describe the overall volume of recent installment activity.

7. Credit Card Balance Data

The Credit Card Balance dataset provides monthly snapshots of applicants' revolving credit card accounts, capturing how balances, limits, and payments evolve over time. Each record corresponds to a specific credit card account at a given month and includes information on outstanding balance, credit limit usage, minimum and total payments, and delinquency indicators.

7.1. EDA Highlights

The Credit Card Balance dataset contains approximately 3.8 million monthly records across 103,558 applicants, where each row represents a monthly snapshot of a credit card account. Contract status information in NAME_CONTRACT_STATUS is highly imbalanced, with Active accounts comprising about 96% of observations, and all other statuses occurring rarely.

7.2. Feature Engineering

7.2.1. Temporal Filtering

Only recent credit card history is retained by restricting the data to the last 12 months using MONTHS_BALANCE. This ensures that engineered features reflect current and relevant revolving credit behavior rather than outdated activity.

7.2.2. New Features

NEW_CRED_UTIL = AMT_BALANCE / AMT_CREDIT_LIMIT_ACTUAL: Measures revolving credit usage relative to available credit. This feature captures how heavily an applicant relies on available credit limits.
CC_COUNT: records the number of recent monthly credit card observations per applicant, capturing the depth of observed revolving credit activity.

7.2.3. Categorical Encoding

The NAME_CONTRACT_STATUS variable is first **rare-encoded**, collapsing infrequent statuses into a single category to reduce sparsity and noise. The cleaned categorical features are then **one-hot encoded**.

7.2.4. Aggregation to Applicant Level (SK_ID_CURR)

After feature construction and encoding, all monthly credit card records are aggregated to the applicant level.

- *Numerical Aggregations:* Maximum and mean statistics are applied to balance, credit limit, payment, drawing, delinquency (SK_DPD, SK_DPD_DEF), and utilization features. These summarize both typical behavior and worst-case exposure across recent months.
- *Categorical Aggregations:* One-hot encoded contract status variables are aggregated using the mean.

8. Modeling

8.1. Final Dataset

After feature engineering each dataset independently, all features are merged into a single applicant-level table (SK_ID_CURR). The Application dataset is used as the base table, and all auxiliary datasets (bureau, previous applications, POS-CASH, installments payments, and credit card balance) are left-joined sequentially. Left joins are used to preserve all applicants present at application time, even if they have no records in external credit history datasets. This ensures the modeling dataset remains aligned with the original prediction task and avoids unintentionally dropping applicants with limited or no prior credit activity. As a result, the final feature table grows from 148 base application features to 386 total features, while retaining 356,249 applicants (see summary table below).

8.1.1. Missing Values in the Final Dataset

The merged dataset contains a substantial number of missing values, particularly in features derived from external credit history sources. This missingness is expected and reflects the structural absence of data rather than data quality issues.

A. Categorical missing values:

- *Application data:* Missing categorical values are explicitly preserved by creating separate categories, as missingness at application time may itself carry a predictive signal.
- *All other datasets:* Missing categories are not encoded as separate levels. In these datasets, missing values primarily indicate the absence of historical records rather than meaningful categorical states. Treating them as explicit categories would add sparsity without introducing additional signal, so nulls are left unmodeled

B. Numerical missing values:

- Numerical features are left as NaN after merging.
- These values are handled at the modeling stage, where tree-based models or consistent imputation strategies can account for missingness.

	Data Source	Rows	Columns	Columns with NaN
1	Application (train/test)	356,249	147	13
2	Bureau + Bureau Balance (agg)	305,811	31	8
3	Previous Applications (agg)	338,857	135	30
4	POS CASH Balance (agg)	240,235	10	0
5	Installments Payments (agg)	252,761	22	12
6	Credit Card Balance (agg)	103,558	40	12
	Total	356,249	386	251

8.2. Logistic Regression (baseline)

8.2.1. Hyperparameter Tuning

Hyperparameter tuning was performed using Bayesian Optimization with Optuna, optimizing the AUC metric as the objective. Due to computational constraints, tuning was conducted on a single training fold only, as running Bayesian optimization across multiple folds significantly increased runtime without proportional gains. The search was run for 20 trials, allowing efficient exploration of the regularization strength and solver configuration.

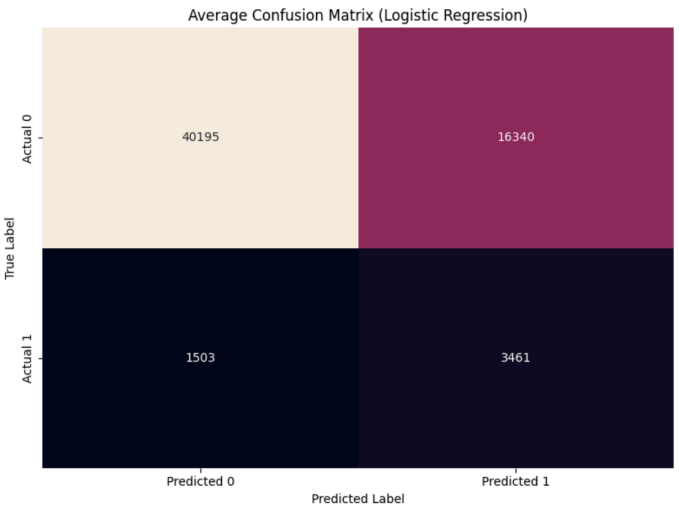
An L2 (ridge) penalty was selected instead of Elastic Net. While Elastic Net can provide additional sparsity, it substantially increases training time for large, high-dimensional feature spaces such as this one. Given the already extensive feature engineering and one-hot encoding, L2 regularization offered a better trade-off between stability, performance, and computational efficiency.

To address class imbalance in the target variable, class weights were applied during training, ensuring that the minority (default) class received higher importance in the loss function. And before model fitting, numerical features were median-imputed to handle missing values, and all features were standardized using z-score normalization.

8.2.2. Tuned Model and Results

Using the best hyperparameters identified during tuning, the Logistic Regression model was then trained and evaluated using 5-fold cross-validation. Each fold followed the same preprocessing pipeline, including median imputation and standard scaling, fitted only on the training portion of the fold to prevent data leakage. Detailed fold-level metrics (AUC and related statistics) are reported below.

Metric	Result
Mean Train AUC	0.77569
Mean Valid AUC	0.77133
Mean PR-AUC	0.24808
Mean Precision (0.5<)	0.17479
Mean Recall (0.5<)	0.69712

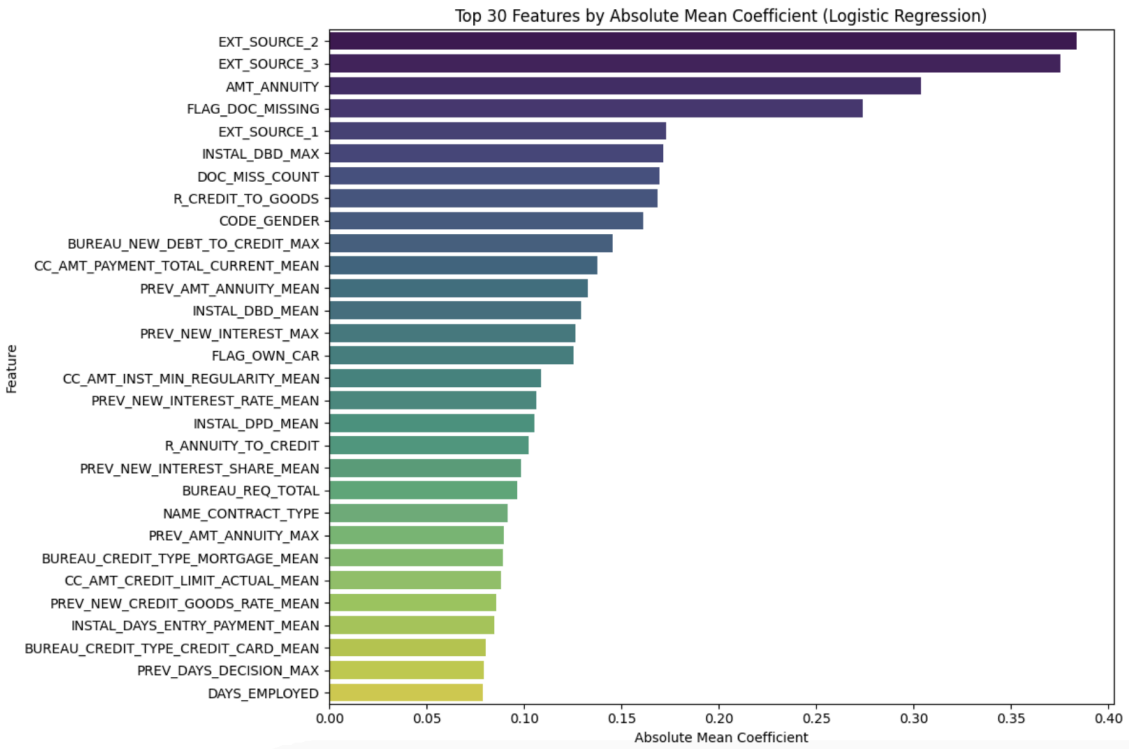


The Logistic Regression model shows stable performance across folds, with mean training AUC (0.776) and validation AUC (0.771) being very close. This indicates no overfitting and good generalization. The PR-AUC (0.248) reflects the strong class imbalance, which is expected in default prediction. At a 0.5 threshold, the model achieves high recall (around 0.70) but lower precision (around 0.17), indicating a deliberate tradeoff. From the confusion matrix, the model favors reducing false negatives (missed defaulters) at the cost of more false positives. In Home Credit risk, this is often preferable, as false negatives lead to direct financial losses, while false positives mainly result in opportunity costs.

8.2.3. Feature Importance

Feature importance is derived from the absolute mean of standardized logistic regression coefficients across CV folds. Several observations can be made from the feature importance histogram below:

- The most influential features (such as EXT_SOURCE_1/2/3, AMT_ANNUITY, debt, and interest-related ratios) align well with earlier correlation results from Section 1.1.4.
- Affordability and repayment burden features, such as AMT_ANNUITY, debt-to-credit ratios, and installment delinquency measures (INSTAL_DBD_MAX/MEAN), rank highly, highlighting the importance of both current financial pressure and historical repayment behavior.
- Several document and application-quality indicators (FLAG_DOC_MISSING, DOC_MISS_COUNT) appear among top predictors, suggesting that incomplete or inconsistent documentation is systematically associated with higher default risk.
- Top features come from multiple sources (application data, bureau history, previous applications, installments, and credit cards), indicating that behavioral and historical credit information adds meaningful predictive power beyond static demographics.



8.3. XGBoost

8.3.1. Feature Selection

Before full hyperparameter tuning, an initial Bayesian Optimization (Optuna) run was performed to obtain a reasonable set of baseline XGBoost parameters. Using these parameters, a 5-fold cross-validated XGBoost model was trained on the full feature set to compute feature importances. Feature importance was calculated from the model's built-in importance scores (gain-based), aggregated across folds.

Based on this ranking, the bottom 30% of least important features were removed. This step was motivated by the very high dimensionality of the merged dataset and aimed to reduce sparsity, noise, and computational cost while retaining the most informative signals. After feature pruning, the dataset was reduced to 272 features.

8.3.2. Hyperparameter Tuning

After feature selection, the XGBoost model was further tuned using Bayesian Optimization with Optuna on the reduced feature set. This stage focused on refining model capacity and regularization after sparsity reduction. Due to computational constraints, hyperparameter tuning was performed on a single training fold, as running Optuna across multiple folds substantially increased runtime. The optimization was run for 20 trials, with AUC as the objective metric.

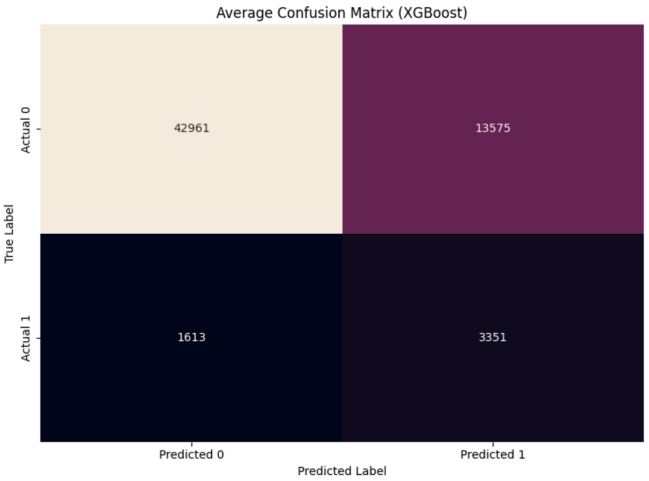
Missing numerical values were left as-is, since XGBoost natively handles missing values by learning optimal default split directions, eliminating the need for explicit imputation. Regularization parameters (L1 alpha and L2 lambda) were included in the search space to control complexity and mitigate overfitting in the high-dimensional feature space.

To address class imbalance in the target variable, the model used class-weighting via scale_pos_weight, ensuring that minority-class (default) observations contributed more strongly to the loss function and improving sensitivity to defaults.

8.3.3. Tuned Model and Results

Using the optimized hyperparameters obtained from Bayesian tuning on the reduced feature set, the final XGBoost model was trained and evaluated using 5-fold cross-validation to obtain stable performance estimates. Below are the results:

Metric	Result
Mean Train AUC	0.86192
Mean Valid AUC	0.79083
Mean PR-AUC	0.28508
Mean Precision (0.5<)	0.19803
Mean Recall (0.5<)	0.67505

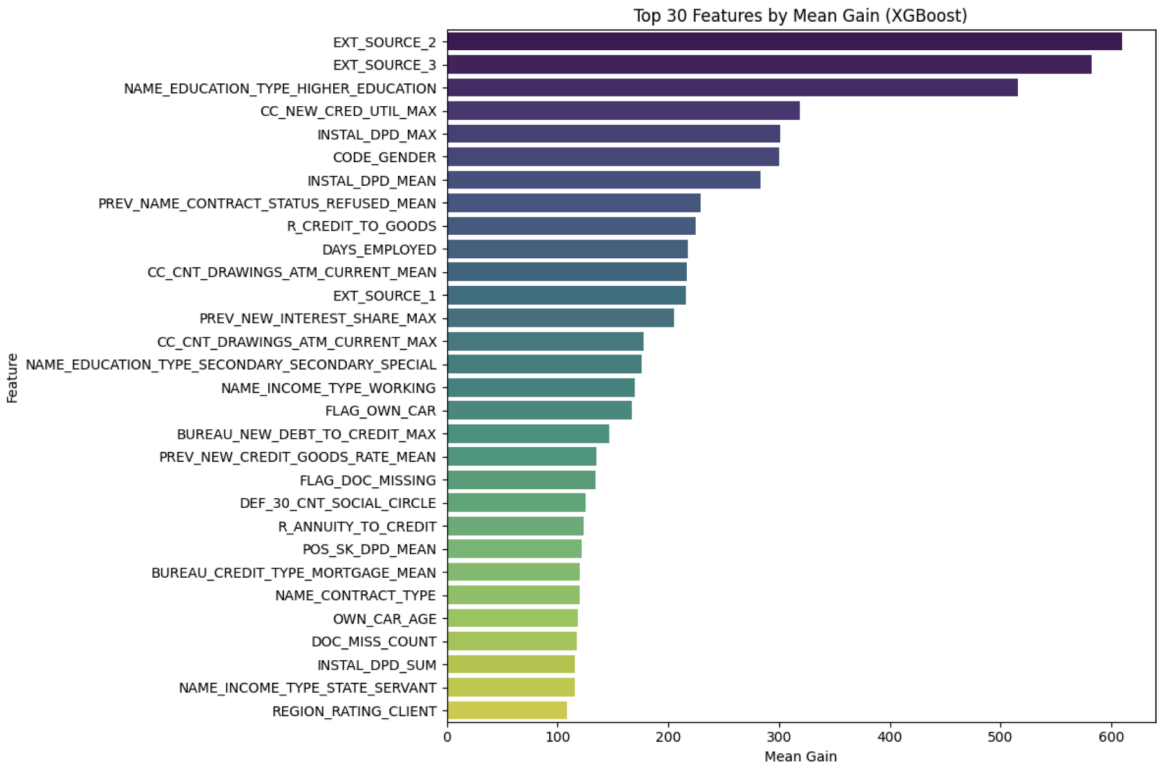


The tuned XGBoost model achieves a higher validation AUC (0.791) than Logistic Regression. The increase in PR-AUC (0.285) further suggests stronger performance on the minority (default) class, which is especially important given the class imbalance. The gap between train AUC (0.862) and validation AUC (0.791) indicates some degree of overfitting, but it remains within a reasonable range for a high-capacity model like XGBoost, particularly given the large, sparse feature space. From the confusion matrix, XGBoost maintains a high recall (around 0.68), meaning it successfully identifies a large share of defaulters, while precision remains modest. This reflects a deliberate trade-off: the model favors reducing false negatives (missed defaults) at the cost of more false positives, which is typically preferable in loan default prediction.

8.3.4. Feature Importance

Feature importance in XGBoost is measured by mean gain, representing the average improvement in the loss function from splits using a feature. Multiple observations can be made from the feature importance histogram below:

- **External bureau scores dominate:** EXT_SOURCE_2 and EXT_SOURCE_3 are the most important features, consistent with their strong negative correlation with the target observed earlier.
- **Behavioral repayment features gain prominence:** Delinquency and utilization features (INSTAL_DPD_MAX/MEAN, POS_SK_DPD_MEAN, CC_NEW_CRED_UTIL_MAX) rank highly. These variables showed weaker linear correlations but are emphasized by XGBoost, indicating important nonlinear effects captured by tree-based models.
- **Previous application signals matter:** Features such as PREV_NAME_CONTRACT_STATUS_REFUSED_MEAN and interest-related ratios align with earlier findings that past credit outcomes and borrowing conditions carry predictive information.
- **Demographics remain secondary:** Variables like education level, income type, employment duration, and car ownership contribute modestly, consistent with their relatively weak correlations with the target.



8.4. LightGBM

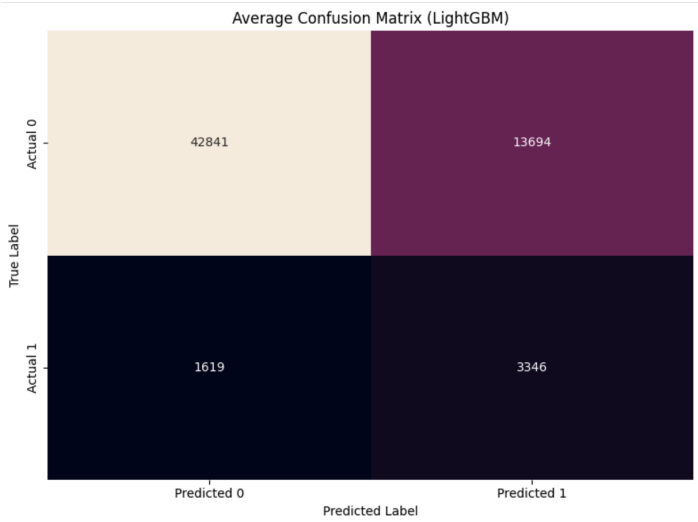
LightGBM followed the same overall modeling workflow as XGBoost: (1) an initial run to obtain baseline hyperparameters and feature importances, (2) feature selection based on LightGBM-specific importance scores, where the least-informative 30% of features were removed to reduce sparsity, (3) Optuna-based hyperparameter tuning on the reduced feature set (272 features), and (4) final evaluation using 5-fold cross-validation.

As in the XGBoost model, numerical missing values were left un-imputed, since LightGBM natively handles missing values during tree splitting. Class imbalance was handled in the same way as XGBoost, by enabling the is_unbalance parameter, which automatically adjusts class weights to give higher importance to the minority (default) class during training.

8.4.1. Results

The gap between mean train AUC (0.875) and mean validation AUC (0.789) is moderate and comparable to XGBoost, indicating good generalization with some expected complexity-driven overfitting. A validation AUC of approximately 0.79 confirms strong ranking ability and places LightGBM on par with XGBoost, clearly outperforming the linear baseline. PR-AUC (around 0.28) remains reasonable given the strong class imbalance and reflects improved minority-class separation compared to logistic regression. At the 0.5 threshold, recall (around 0.67) is relatively high, while precision (around 0.20) remains low. This reflects a deliberate bias toward catching defaulters, which is appropriate in credit risk settings.

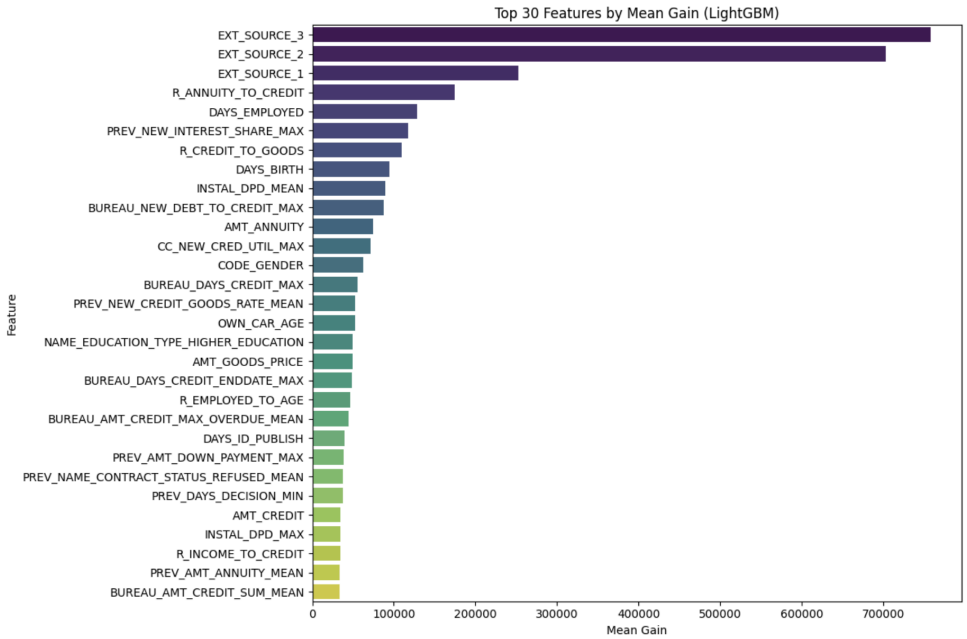
Metric	Result
Mean Train AUC	0.87496
Mean Valid AUC	0.78934
Mean PR-AUC	0.28223
Mean Precision (0.5<)	0.19644
Mean Recall (0.5<)	0.67392



8.4.2. Feature Importance

In LightGBM, feature importance is computed based on gain, which measures the total reduction in the training loss brought by splits on a given feature, averaged across all trees. Here are a couple of observations based on the feature importance histogram below:

- **Strong concentration of importance:** A small set of features (notably EXT_SOURCE_1/2/3, annuity and credit ratios, employment and delinquency signals) dominates the total gain. These variables consistently produce the largest reductions in loss when used for splits, indicating they carry the most predictive signal.
- **Long tail of low-importance features:** The remaining features contribute marginal, incremental improvements. While individually weak, they can still improve performance collectively by refining decision boundaries or interacting with stronger features.



9. Limitations and Future Work

While the proposed models achieved competitive performance, they did not surpass the top competition results. This gap can be explained by several practical and methodological factors rather than fundamental modeling flaws.

Submission and Description	Private Score	Public Score
submission_lbgm.csv Complete (after deadline) · 15s ago	0.78636	0.78970
submission_logreg.csv Complete (after deadline) · 30s ago	0.76503	0.76874
submission_xgb.csv Complete (after deadline) · 1m ago	0.78854	0.79086

Time and Experimentation Constraints

The original competition spanned approximately 3.5 months, allowing top teams to iteratively experiment with advanced feature engineering, extensive validation strategies, and ensemble refinement. In contrast, this project was completed within a significantly shorter timeframe, which limited the depth of experimentation, ablation studies, and fine-grained tuning that could be performed.

Feature Engineering Opportunities

Although extensive cross-dataset feature engineering was implemented, further gains could likely be achieved through:

- **Temporal and lag-based features**, especially for behavioral datasets (installments, POS, credit card balances).
- **Dimensionality reduction or feature grouping** (ex: target-encoding variants, feature hashing, or PCA-style aggregation for high-cardinality signals).
- **More domain-specific features**, such as payment regularity trends, rolling delinquency patterns, or stability measures over time, rather than static aggregates.

Model Complexity and Ensembling

The project focused on strong baseline and tree-based models (Logistic Regression, XGBoost, LightGBM). Other approaches were not fully explored due to time and computational constraints. Top competition solutions typically rely on:

- **Ensemble methods** (blending or stacking multiple gradient boosting models with different feature subsets).
- **Neural networks or hybrid architectures**, particularly to capture nonlinear interactions not fully exploited by tree depth alone.

Missing Value Treatment

While missing values were handled in a principled and model-appropriate way (median imputation for linear models; native handling for tree-based models), performance could potentially improve with:

- **Feature-specific imputation strategies**, where missingness carries semantic meaning (ex: zero vs. “not applicable” vs. truly missing).
- Explicit modeling of missingness indicators beyond the application dataset.

Validation and Overfitting Control

Hyperparameter tuning was intentionally limited (ex: Bayesian optimization on a single fold) to keep runtime manageable. Although cross-validated results show stable generalization, additional gains may come from:

- More extensive nested cross-validation during tuning.
- More aggressive regularization schedules.
- Targeted overfitting analysis on high-variance features.